Kasra Mirnobahar

0009-0001-2363-6381 | Kasramirnobahar.github.io

Kasra.Mirnobahar@glasgow.ac.uk
Kasranobahar@gmail.com

Research Interests

I am interested in how digital health, decision-making, and behaviour interact to shape mental health outcomes. My work focuses particularly on social media, using experimental tasks, computational and reinforcement learning modelling, real-world data, data donation, and neuroimaging methods to understand how reward systems shape behaviour. My long-term goal is to better understand digital well-being and design interventions and applied frameworks that can inform public policy and improve related mental health outcomes.

Education

MSc Behavioural Sciences, University of Glasgow — Sep 2024 - Sep 2025
GPA: Merit
Dissertation: The Impact of Reward Timing Predictability on Checking Behaviour in Social Media (Graded A3)

Bachelor of Psychology, University of Social Welfare and Rehabilitation Sciences — Aug 2019 - Oct 2023
GPA: 17.58 (Out of 20) - 3.71/4 - First Class Honours
Dissertation: The relationship between Poly-Victimisation and Behavioural Problems Among Individuals with Psychiatric Disorders (Graded A1)

Awards & Funding

Behavioural Sciences Scholarship, University of Glasgow (2024) – awarded for the MSc in Behavioural Sciences

Publications

- **In Preparation:** The Social Media Skinner Box: Uncertain Reward Invigorates Social Media Checking
- Pourmohamadreza-Tajrishi M, **Mirnobahar K**, Mohammadi Bytamar J. Assessing Childhood Trauma: The Iranian Validation of the Juvenile; Polyvictimizaion Questionnaire and Its Link to Psychiatric Symptoms. Int J High Risk Behav Addict. 2025; 14 (3): e163153. https://doi.org/10.5812/ijhrba-163153
- **Mirnobahar K**, Pourmohamadreza-Tajrishi M, Mohammadi Bytamar J. Impact of Childhood Poly-victimization on Impulsivity and Behavioral Problems in Iranian Adults: A Regression Analysis.Iran J Psychiatry Behav Sci.2025;19(3):e159188.https://doi.org/10.5812/ijpbs-159188.
- **Under review:** The Impact of FoMO and Belonging on Problematic Social Media Use: Social Isolation as a Mediating Factor. Under review at Scientific Reports.

Research Experiences and Projects

Research Collaborator – How Social Media makes us feel? — Aug 2026 - Ongoing
Collaboration with David Grüning, Georgia Turner, Wouter van den Bos, Myranda.ai, and How We Feel
Max-Plank Institute, University of Cambridge, University of Amsterdam,

- Using real-data from social media combined with EMA to investigate how emotion and social media usage shape individuals' online behaviour

Research Collaborator – Causal Inference in Psychosis EHR — June 2026 - Ongoing
Collaborating with Dr. Samuel Leighton, University of Glasgow, School of Health & Well-Being & Rajeev Krishnadas University of Cambridge

- Developing a target trial emulation protocol using longitudinal electronic health record data from patients with psychosis from NHS Greater Glasgow and Clyde (Esteem Project – SafeHaven)
- Data Cleaning, Applying causal inference methods, including target trial emulation and survival analysis, with planned analyses incorporating inverse probability weighting, adjustment for baseline and time-varying clinical confounding, censoring mechanisms, and sensitivity analyses.
- Contributing to related work on antipsychotic medication and obesity in people with psychosis

Primary Researcher - Social Media and Uncertainty — June 2025 - Sep 2025
University of Glasgow, Adam Smith Business School, Supervisors: Federica Farolfi
Georgia Turner, MRC Cognition and Behavioural Sciences, University of Cambridge

- Independently conceptualised and designed a study examining how reward timing predictability shapes social media checking behaviour
- Designed a multi-component experimental task in Gorilla.sc, including a simulated social media environment feed, where participants can explore feed and upload their own posts and receive likes
- Led ethical approval process, recruited and screened participants, and coordinated all data collection logistics, recruited 110 participants, ensured data integrity throughout collection and analysis in line with ethical research standards
- Conducted full quantitative analysis in R; wrote and edited the dissertation manuscript
- Full analysis plan, pipeline, and updated version of task materials openly available on GitHub

Research Experiences (continued)

Research Assistant (EEG/fMRI) - Decision-making and Metacognition of Confidence — June 2025 – March 2026
Philiastides Lab, University of Glasgow, School of Psychology and Neuroscience – Collaborator: Dr. Gina Brunner

- Contributing to data collection including EEG and fMRI Setup on perceptual decision-making and metacognition for 45 subjects
- Supporting experimental setup, participant testing, and data acquisition
- Assisting with data quality control, EEG data preprocessing using MATLAB and EEGLab

Research Assistant (EEG/ECG) - Decision-making, Learning and Cardiac Cycle — Feb 2026 – June 2026
Philiastides Lab, University of Glasgow, School of Psychology and Neuroscience – Collaborator: Dr. Hamed Ghane

- Contributed to EEG/ECG data collection and participant recruitment across 30 testing sessions
- Supporting experimental setup, participant testing, and data acquisition
- Assisting with data quality control, data cleaning and analysis using Python

Research Assistant - Social Media Problematic Usage — Feb 2023 - Sep 2023
University of Social Welfare & Rehabilitation Sciences Department of Clinical Psychology

- Coordinated participant recruitment, screening, and ongoing engagement across study sessions for both clinical and non-clinical samples
- Conducted online and in-person data collection for studies on problematic social media use,
- Organised, cleaned, and managed datasets; performed data entry, wrangling, and regular quality checks for statistical analysis
- Contributed to manuscript drafting, revision, and editing for peer-reviewed publication

Research Assistant - Cognitive HTKS Task among Children via Schools — June 2022 – Feb 2023
University of Social Welfare & Rehabilitation Sciences Department of Psychology and Education of Exceptional Children

- Coordinated participant recruitment from school settings for children aged 5–8, liaising directly with school administrators and teachers to obtain institutional access and schedule data collection visits
- Managed the informed consent and assent process with parents, guardians, ensuring all procedures met ethical and safeguarding standards
- Administered multiple cognitive and behavioural task in classroom environments across multiple school visits, adapting delivery to child-appropriate communication
- Organised, coded, and maintained structured participant records and behavioural datasets in SPSS, ensuring accuracy and completeness across multiple data collection sessions
- Led the full ethical approval process with governmental and educational authorities, including risk assessment for research involving young children

Primary Researcher - Childhood poly-victimisation, Behavioural Problems & Impulsivity (BSc Dissertation) — March 2023 - Sep 2023
University of Social Welfare & Rehabilitation Sciences Department of Clinical Psychology

- Independently conceptualised and designed the study; led the full ethical approval process
- Conducted data from adult samples (N = 600); performed regression analysis and psychometric validation in SPSS and R
- Manuscript writing and Editing; Findings published in two peer-reviewed journals

Invited Presentations, Conferences

- **Conference poster – Research Proposal - Person-Specific Digital Phenotyping of Social Media Use and Digital Well-Being**
  Digital Mental Health Conference 2026; University of Cambridge • Dec 2026 (Upcoming)

- **Conference poster - Does Uncertainty Drive Compulsive Checking? Causal Evidence from a Simulated Social Media Task** • July 2026
  BPS Cyberpsychology Annual Conference 2026; York St John University

- **Social Media and Uncertainty: How Reward Timing Predictability Shapes Checking Behaviour** • Nov 2025
  University of Amsterdam, Connected Minds Lab

Working Experience

Researcher; Jahan Majd Basir Co. — Aug 2023 - Dec 2023
Conducted cross-cultural research on social, cultural, religious, and belief systems across multiple regions
Contributed to data collection and synthesis for applied research projects

Science Communicator; Varan Psychology School — Oct 2021 - July 2022
Produced accessible psychology content across clinical, cognitive, and neuroscience topics for an educational platform; coordinated workshops, guest lectures, and professional development sessions.

Volunteering Experiences

Peer-Reviewer, InFrame Project, University of Glasgow, Edinburgh and St Andrews — Jan 2025

- Reviewed and evaluated eight research proposals for scientific merit, feasibility, and alignment with project objectives
- Provided structured feedback to applicants to improve methodological clarity and impact
- Collaborated with a multidisciplinary review panel to support research quality and inclusivity

Suicide Crisis Counsellor, Iran Hotline Suicide Prevention (NGO) — June 2023 – June 2024

- Provided crisis support and intervention for individuals experiencing acute suicidal ideation, following structured clinical safeguarding protocols including escalation procedures
- Developed skills in sensitive, trauma-informed communication with vulnerable individuals in distress; maintained professional boundaries and accurate case documentation

Administrative Coordinator, Counselling centre; University of Social Welfare and Rehabilitation — Jan 2023 - July 2023

Organised and coordinated counselling workshops and academic events; managed scheduling, logistics, and stakeholder communications

Skills

Statistical and computational methods
Regression analysis; Bayesian modelling; survival analysis; causal inference; target trial emulation; inverse probability weighting; longitudinal data analysis; experimental and survey design; computational modelling; reinforcement learning modelling; data cleaning, wrangling, and visualisation.

Programming and software
R; Python; MATLAB/EEGLAB; SPSS; Gorilla.sc; FSL; PsychoPy; GitHub.

Neuroimaging and psychophysiology
EEG setup and preprocessing; fMRI setup and data acquisition; ECG data collection; physiological data quality checks.

Languages

Persian: Native **English:** Fluent **German:** Basic